

CustomerIQ — Product PRD

From ML risk scoring to Power BI intelligence, AI recommendations, and a functional web application

Document purpose

This PRD defines the next product phase of CustomerIQ. The existing project has progressed from raw retail transaction data through customer-level feature engineering, churn modeling, validation, and customer risk scoring. The next objective is to turn those analytical outputs into a business-usable intelligence product.

Core product principle: do not build a decorative website about CustomerIQ. Build CustomerIQ itself — a usable interface that connects customer data, analytics, ML risk, segmentation, and actionable recommendations.

1. Product Vision

CustomerIQ will transform transactional customer data into customer-level risk intelligence that a marketer or business user can understand and act upon.

Target product flow: Customer Data → SQL/Data Layer → ML → Risk Score → Customer Segment → Power BI Analytics → AI Recommendations → CustomerIQ Web App

2. Current State Before This Product Phase

Stage	Completed output
Days 1–2	Raw retail data cleaned and validated; processed transaction dataset created.
Days 3–4	Customer behavior, churn methodology, descriptive retention analysis and customer-level feature dataset.
Day 5	Logistic Regression baseline; ROC-AUC 0.7731; predictions exported.
Day 6	Logistic Regression vs Random Forest vs Gradient Boosting; Gradient Boosting strongest candidate.
Day 7	3-fold validation, leakage checks, confusion matrices, ROC/PR, threshold and feature analysis.
Day 8	3,705 customers converted into HIGH/MEDIUM/LOW risk scores.
Day 9	SQLite/SQL data layer, business queries and Power BI-ready dataset created.

3. Product Objective

Move CustomerIQ from an analytical project into a demonstrable business intelligence and customer-retention product. The product must make model outputs understandable, searchable, visual, and actionable.

Primary objectives:

- Build a professional Power BI analytics layer.
- Create an AI recommendation layer grounded in customer risk and behavioral signals.
- Build a lightweight CustomerIQ web application around the proven analytical logic.
- Keep the MVP explainable, auditable, and free from unsupported claims.

4. Target Users

Primary: marketers, CRM/retention teams, business analysts and managers who need to prioritize customers.

Secondary: recruiters, interviewers, teachers and portfolio reviewers evaluating the project's analytics, ML and product-building capability.

5. Power BI Requirements

Power BI is the first user-facing layer to build because it validates whether the model outputs can answer real business questions before web-app productization.

Dashboard component	Purpose
KPI cards	Total customers, high-risk customers, risk distribution and relevant customer-value measures.
Risk distribution	Show HIGH/MEDIUM/LOW customer segments clearly.
High-risk customer table	Prioritize customers for investigation or intervention.
Risk vs customer value	Identify valuable customers who may deserve higher retention priority.
Behavioral drivers	Connect model/risk outputs to customer behavior.
Interactive filters	Allow users to explore customer segments and risk groups.
Model-performance view	Communicate model quality without hiding limitations.

6. AI Recommendation Layer

The recommendation system should not initially pretend that generative AI itself predicts churn. The validated ML model produces risk; deterministic business rules translate risk and customer context into recommended action categories. A GenAI layer may then convert those structured signals into clear, marketer-friendly explanations.

Example rule: HIGH risk + high customer value → prioritize a targeted win-back campaign.

Example output: risk level, key behavioral signals, recommended action, reason, and suggested priority.

Guardrail: recommendations must be presented as decision support, not guaranteed outcomes. The product should expose the evidence behind a recommendation.

7. CustomerIQ Web Application — MVP

The web app is the product layer built after the Power BI analytical layer is working. It should be lightweight and functional rather than overloaded with features.

Screen / feature	MVP requirement
Dashboard	Overall customer count, risk KPIs, distribution and priority overview.
Customer list	Search/filter customers by risk and relevant business attributes.
Customer profile	Show risk probability, segment, behavioral features and relevant drivers.
Recommendation panel	Show recommended retention action and the reasoning behind it.
Risk prioritization	Make high-value/high-risk customers easy to identify.
Data connection	Read from the structured SQLite/CSV analytical layer.
Responsive UI	Clean, professional interface suitable for portfolio demonstration.

8. Technical Architecture

Data: raw retail data → cleaned transaction data → customer-level analytical data

Database: SQLite + SQL business queries

ML: scikit-learn; Gradient Boosting currently the strongest candidate

Risk: customer-level churn probability and HIGH/MEDIUM/LOW segmentation

BI: Power BI dashboard and interactive analysis

AI: rule-based recommendation engine followed by optional natural-language generation

Application: lightweight web application exposing dashboard, customer profiles and recommendations

9. AI-Agent Division of Work

AI agent should handle: repetitive Python/SQL/frontend code generation, file creation, schema checks, data validation, API scaffolding, repetitive UI components, automated tests, documentation drafts and execution/verification scripts.

User should handle: understanding the data, validating business assumptions, choosing what a marketer needs, interpreting results, reviewing AI-generated code, testing the product, making design decisions, and presenting the project.

The objective is not to avoid technical understanding. It is to avoid spending the majority of project time manually typing repetitive implementation code.

10. Development Roadmap — Days 10–17

Days	Focus	Definition of done
10–11	Power BI	Working dashboard answering customer-risk and business-priority questions.
12	Recommendation logic	Deterministic rules producing explainable action categories.
13–14	Web app MVP	Dashboard + customer list + customer profile + recommendation view.
15–16	Integration & polish	End-to-end flow tested; UI, data and recommendation outputs consistent.
17	Portfolio & presentation	Final demo, architecture, README, screenshots, limitations and presentation story.

11. Success Criteria

The MVP is successful if a new user can understand what CustomerIQ does, identify high-risk customers, inspect why they are prioritized, view the relevant customer behavior, and receive a clear recommended next action — without opening a notebook.

Technical success:

- Existing Days 1–9 remain reproducible and source data remains protected.
- Power BI uses the structured Day 9 dataset.
- Web app uses the same underlying analytical definitions as the validated ML pipeline.
- Risk scores and segments remain consistent across CSV, Power BI and application views.
- AI recommendations are traceable to structured inputs.

12. Out of Scope for the MVP

Deep learning, complex agent architecture, production-scale cloud infrastructure, advanced survival analysis, large-scale real-time streaming, and unnecessary model experimentation are not required for the 15–17 day MVP unless a later iteration demonstrates a clear business need.

13. Product Positioning

CustomerIQ is not just a churn model. It is a customer intelligence workflow that connects data, machine learning, business intelligence and decision support.

The portfolio story should emphasize the progression: raw data → customer behavior → churn methodology → ML → validation → risk scoring → SQL → Power BI → recommendations → product.

14. Final Deliverable

By the end of the product phase, CustomerIQ should be demonstrable as a coherent end-to-end system rather than a collection of disconnected notebooks. The final presentation should show both the analytical credibility of the model and the practical value of converting predictions into customer-retention decisions.